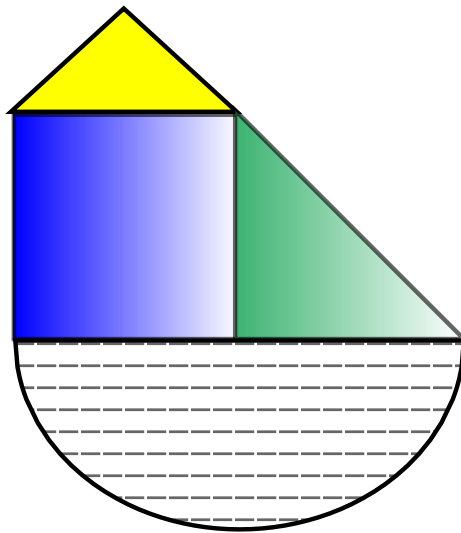


8.2 Area of Composite Figures

A **composite figure** is formed by combining simple shapes. You can calculate the total area of a composite figure by adding or subtracting areas.

My creation:



What Formulas do we Know?

Area of a rectangle: $length \times width$

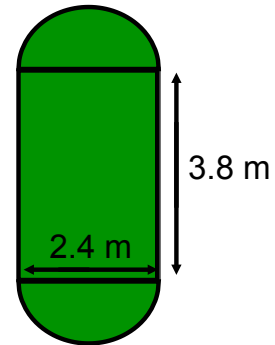
Area of a triangle: $\frac{1}{2}bh$

Area of a circle: πr^2

Perimeter of a circle: $2\pi r$ or πd

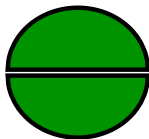
Area of Composite Figures

1) Find the area of the following composite figure:

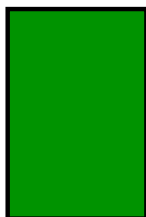


Step 1: Identify the shapes that make up the figure

2 semi-circles
= 1 circle



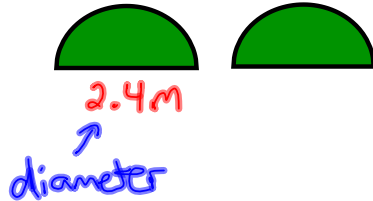
1 rectangle



Step 2: Find the area of each shape

Area of 2 semicircles:

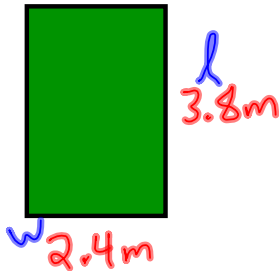
= 1 circle



$$\begin{aligned} A &= \pi r^2 \\ &= \pi (1.2)^2 \\ &= 4.5 \text{ m}^2 \end{aligned}$$

$$\begin{aligned} \text{radius} &= \frac{\text{diameter}}{2} \\ &= \frac{2.4}{2} \\ &= 1.2 \text{ m} \end{aligned}$$

Area of rectangle:



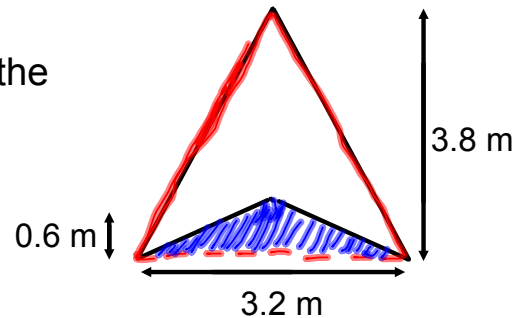
$$\begin{aligned} A &= l \times w \\ &= 3.8(2.4) \\ &= 9.1 \text{ m}^2 \end{aligned}$$

Step 3: Calculate the total area

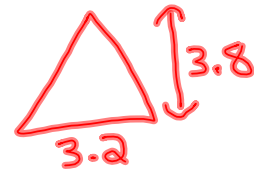
$$\begin{aligned} A_{\text{total}} &= A_{\text{circle}} + A_{\text{rectangle}} \\ &= 4.5 + 9.1 \\ &= 13.6 \text{ m}^2 \end{aligned}$$

2) Find the area of the hang glider sail:

Step 1: Determine how we can find the area of this sail



Step 1: Determine the area of the large triangle



Step 2: Determine the area of the small triangle that has been "cut out" from the bigger triangle



Step 3: Find the difference in their areas

Step 2: Calculate the area of each figure

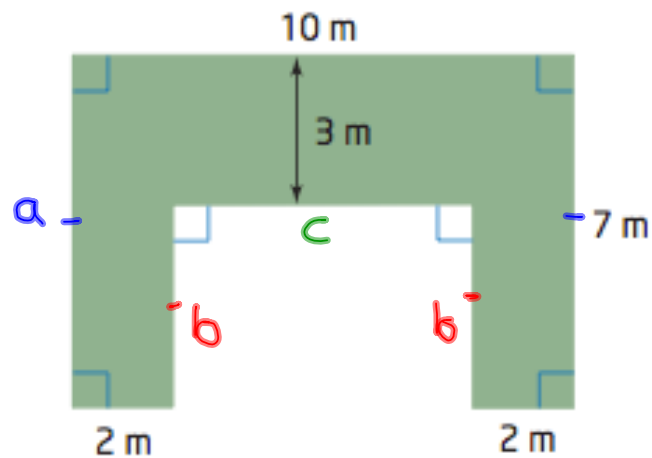
$$A_{\text{large triangle}} = \frac{bh}{2} = \frac{3.2(3.8)}{2} = 6.08 \text{ m}^2$$

$$A_{\text{small triangle}} = \frac{bh}{2} = \frac{3.2(0.6)}{2} = 0.96$$

Step 3: Find the total area

$$\begin{aligned} A_{\text{total}} &= A_{\text{large}} - A_{\text{small}} \\ &= 6.08 - 0.96 \\ &= 5.12 \text{ m}^2 \end{aligned}$$

3) a) Find the length of the unknown sides for the following object:



$$a = 7 \text{ m}$$

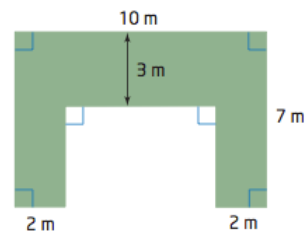
$$\begin{aligned} b &= 7 - 3 \\ &= 4 \text{ m} \end{aligned}$$

$$\begin{aligned} c &= 10 - 2 - 2 \\ &= 6 \text{ m} \end{aligned}$$

b) Find the perimeter of the object:

$$P = 10 + 7 + 2 + 4 + 6 + 4 + 2 + 7$$

$$= 42 \text{ m}$$



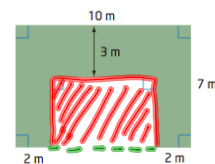
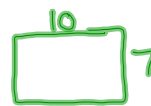
Note: only add the outside sides of the object!

c) Find the area of the object:

$$A_{\text{large rectangle}} = l \times w$$

$$= 10(7)$$

$$= 70 \text{ m}^2$$



$$A_{\text{small rectangle}} = l \times w$$

$$= 6(4)$$

$$= 24 \text{ m}^2$$



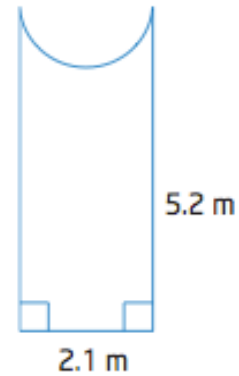
$$A_{\text{total}} = A_{\text{large}} - A_{\text{small}}$$

$$= 70 - 24$$

$$= 46 \text{ m}^2$$

- 4) A hotel is remodelling its outdoor entrance area. The new design includes a tile walkway leading to a semicircular fountain.

a) What steps would you take to find the area of the walkway?



Step 1: Find area of rectangle

Step 2: Find area of semicircle that is "cut out" of rectangle

Step 3: Find difference in areas

b) Calculate the area of the walkway (round to the nearest tenth)

$$\begin{aligned} A_{\text{rectangle}} &= l \times w \\ &= 2.1(5.2) \\ &= 10.92 \text{ m}^2 \end{aligned}$$

$$\begin{aligned} A_{\text{semi-circle}} &= \frac{A_{\text{circle}}}{2} = \frac{\pi r^2}{2} \\ &= \frac{\pi (1.05)^2}{2} \\ &= \frac{3.46 \text{ m}^2}{2} \\ &= 1.73 \text{ m}^2 \end{aligned}$$

$$\text{Note: } r = \frac{2.1}{2} = 1.05$$

$$\begin{aligned} A_{\text{total}} &= A_{\text{rectangle}} - A_{\text{semicircle}} \\ &= 10.92 - 1.73 \\ &= 9.19 \text{ m}^2 \end{aligned}$$

c) The walkway will have a border in a different colour of tile. Calculate the perimeter of the walkway. (Round to the nearest tenth of a meter)

$$\begin{aligned}\text{Perimeter of semi-circle} &= \frac{\text{circumference}}{2} = \frac{\pi d}{2} \\ &= \frac{\pi(2.1)}{2} \\ &= 3.3 \text{ m}\end{aligned}$$

$$\begin{aligned}\text{Total Perimeter of walkway} &= 5.2 + 5.2 + 2.1 + 3.3 \\ &= 15.8 \text{ m}\end{aligned}$$

Homework:

Complete worksheet

